

THE CELESTIAL SPHERE

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FOR CENTURIES, humans have known that stars lie at different distances from Earth. However, when recording the positions of stars in the sky, it is convenient to

pretend that they are all stuck to the inside of a sphere that surrounds Earth. The idea of this sphere also helps astronomers understand how their location on Earth, the time of night, and the time of year affect what they see in the night sky.

THE SKY AS A SPHERE

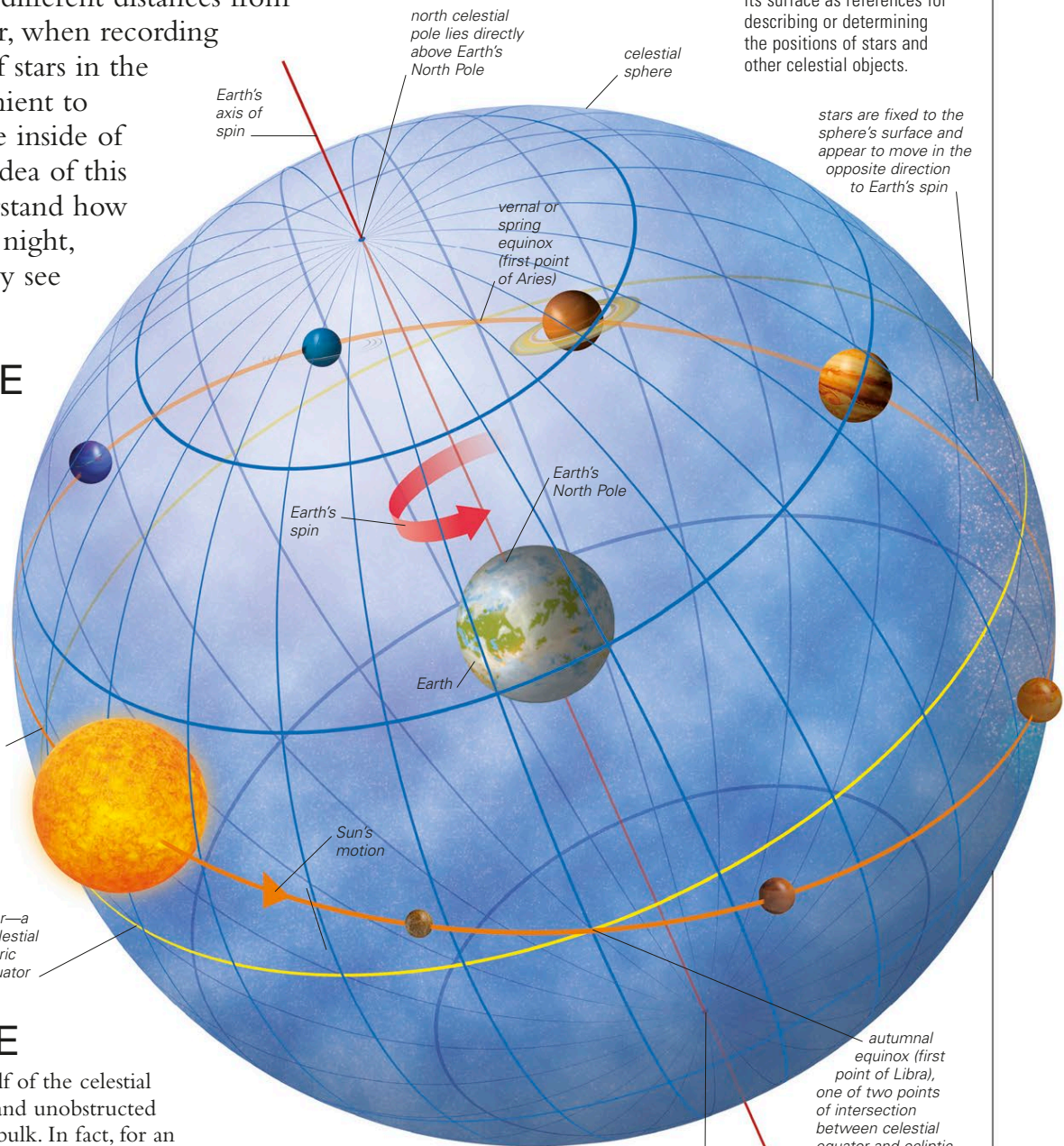
To an observer on Earth, the stars appear to move slowly across the night sky. Their motion is caused by Earth's rotation, although it might seem that the sky is spinning around our planet. To the observer, the sky can be imagined as the inside of a sphere, known as the celestial sphere, to which the stars are fixed and relative to which the Earth rotates. This sphere has features related to the real sphere of the Earth. It has north and south poles, which lie on its surface directly above Earth's North and South Poles, and it has an equator (the celestial equator), which sits directly above Earth's equator. The celestial sphere is like a celestial version of a globe—the positions of stars and galaxies can be recorded on it, just as cities on Earth have their positions of latitude and longitude on a globe.

the Sun and planets are not fixed on the celestial sphere, but move around on or close to a circular path called the ecliptic

celestial equator—a circle on the celestial sphere concentric with Earth's equator

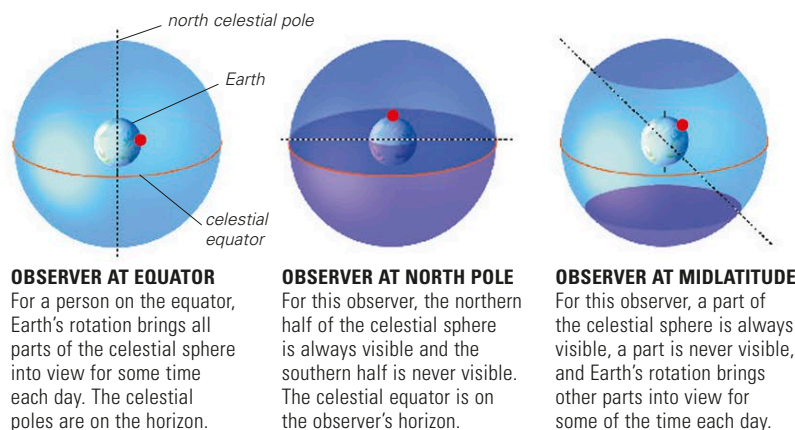
IMAGINARY GLOBE

The celestial sphere is purely imaginary, with a specific shape but no precise size. Astronomers use exactly defined points and curves on its surface as references for describing or determining the positions of stars and other celestial objects.



EFFECTS OF LATITUDE

An observer on Earth can view, at best, only half of the celestial sphere at any instant (assuming a cloudless sky and unobstructed horizon). The other half is obscured by Earth's bulk. In fact, for an observer at either of Earth's poles, a specific half of the celestial sphere is always overhead, while the other half is never visible. For observers at other latitudes, Earth's rotation continually brings new parts of the celestial sphere into view and hides others. This means, for example, that over the course of a night, an observer at a latitude of 60°N or 60°S can see up to three-quarters of the celestial sphere for at least some of the time, and an observer at the equator can see every point on the celestial sphere at some time.



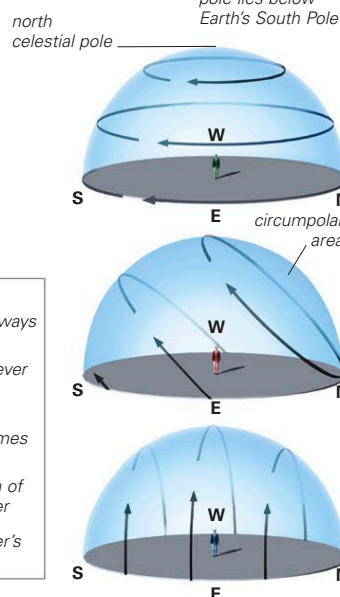
OBSERVER AT EQUATOR
For a person on the equator, Earth's rotation brings all parts of the celestial sphere into view for some time each day. The celestial poles are on the horizon.

OBSERVER AT NORTH POLE
For this observer, the northern half of the celestial sphere is always visible and the southern half is never visible. The celestial equator is on the observer's horizon.

OBSERVER AT MIDLATITUDE
For this observer, a part of the celestial sphere is always visible, a part is never visible, and Earth's rotation brings other parts into view for some of the time each day.

KEY

- stars always visible
- stars never visible
- stars sometimes visible
- position of observer
- observer's horizon



MOTION AT NORTH POLE

At the poles, all celestial objects seem to circle the celestial pole directly overhead. The motion is counter-clockwise at the North Pole and clockwise at the south.

MOTION AT MIDLATITUDE

At midlatitudes, most stars rise in the east, cross the sky obliquely, and set in the west. Some (circumpolar) objects never rise or set, but circle the celestial pole.

MOTION AT EQUATOR

At the equator, stars and other celestial objects appear to rise vertically in the east, move overhead, and then fall vertically and set in the west.